

Newman-Penrose Formalism Calculations

Stephani et al., Equation (31.57)

Metric

$$ds^2 = \left(-2v^2 \left(\frac{\partial f_x}{\partial x} \right)^2 - 4v^2 \frac{\partial f_x}{\partial x} \frac{\partial f_x}{\partial y} - 2v^2 \frac{\partial f_x^2}{\partial y} - 2v^2 \left(\frac{\partial f_y}{\partial x} \right)^2 - 4v^2 \frac{\partial f_y}{\partial x} \frac{\partial f_y}{\partial y} - 2v^2 \frac{\partial f_y^2}{\partial y} - 2H^0(u, x, y) \right) du \otimes du - du \otimes dv - dv \otimes du \\ + \left(\frac{1}{f_x(x, y)^4 + f_y(x, y)^4 + 2(f_x(x, y)^2 + 1)f_y(x, y)^2 + 2f_x(x, y)^2 + 1} \right) dx \otimes dx + \left(\frac{1}{f_x(x, y)^4 + f_y(x, y)^4 + 2(f_x(x, y)^2 + 1)f_y(x, y)^2 + 2f_x(x, y)^2 + 1} \right) dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, v, x, y)$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(v^2 \frac{\partial}{\partial x} f_x(x, y)^2 + 2v^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) + v^2 \frac{\partial}{\partial y} f_x(x, y)^2 + v^2 \frac{\partial}{\partial x} f_y(x, y)^2 + 2v^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) + v^2 \frac{\partial}{\partial y} f_y(x, y)^2 + H^0(u, x, y) \right) du + (1) dv$$

$$m = \left(\frac{\sqrt{\frac{1}{2}}}{\sqrt{f_x(x, y)^4 + 2f_x(x, y)^2 f_y(x, y)^2 + f_y(x, y)^4 + 2f_x(x, y)^2 + 2f_y(x, y)^2 + 1}} \right) dx + \left(\frac{i\sqrt{\frac{1}{2}}}{\sqrt{f_x(x, y)^4 + 2f_x(x, y)^2 f_y(x, y)^2 + f_y(x, y)^4 + 2f_x(x, y)^2 + 2f_y(x, y)^2 + 1}} \right) dy$$

$$\bar{m} = \left(\frac{\sqrt{\frac{1}{2}}}{\sqrt{f_x(x, y)^4 + 2f_x(x, y)^2 f_y(x, y)^2 + f_y(x, y)^4 + 2f_x(x, y)^2 + 2f_y(x, y)^2 + 1}} \right) dx \\ + \left(-\frac{i\sqrt{\frac{1}{2}}}{\sqrt{f_x(x, y)^4 + 2f_x(x, y)^2 f_y(x, y)^2 + f_y(x, y)^4 + 2f_x(x, y)^2 + 2f_y(x, y)^2 + 1}} \right) dy$$

Contravariant components

$$l = (-1) \, dv$$

$$n = (-1) \, du + \left(v^2 \frac{\partial}{\partial x} f_x(x, y)^2 + 2 v^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) + v^2 \frac{\partial}{\partial y} f_x(x, y)^2 + v^2 \frac{\partial}{\partial x} f_y(x, y)^2 + 2 v^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) + v^2 \frac{\partial}{\partial y} f_y(x, y)^2 + H^0(u, x, y) \right) dv$$

$$m = \left(\frac{1}{2} \sqrt{2} f_x(x, y)^2 + \frac{1}{2} \sqrt{2} f_y(x, y)^2 + \frac{1}{2} \sqrt{2} \right) dx + \left(\frac{1}{2} i \sqrt{2} f_x(x, y)^2 + \frac{1}{2} i \sqrt{2} f_y(x, y)^2 + \frac{1}{2} i \sqrt{2} \right) dy$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} f_x(x, y)^2 + \frac{1}{2} \sqrt{2} f_y(x, y)^2 + \frac{1}{2} \sqrt{2} \right) dx + \left(-\frac{1}{2} i \sqrt{2} f_x(x, y)^2 - \frac{1}{2} i \sqrt{2} f_y(x, y)^2 - \frac{1}{2} i \sqrt{2} \right) dy$$

Directional Derivatives

$$D = -\partial_v$$

$$\Delta = -\partial_u + v^2 \frac{\partial}{\partial x} f_x(x, y)^2 + 2 v^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) + v^2 \frac{\partial}{\partial y} f_x(x, y)^2 + v^2 \frac{\partial}{\partial x} f_y(x, y)^2 + 2 v^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) + v^2 \frac{\partial}{\partial y} f_y(x, y)^2 + H^0(u, x, y) \, \partial_v$$

$$\delta = \frac{1}{2} \sqrt{2} f_x(x, y)^2 + \frac{1}{2} \sqrt{2} f_y(x, y)^2 + \frac{1}{2} \sqrt{2} \partial_x + \frac{1}{2} i \sqrt{2} f_x(x, y)^2 + \frac{1}{2} i \sqrt{2} f_y(x, y)^2 + \frac{1}{2} i \sqrt{2} \partial_y$$

$$\bar{\delta} = \frac{1}{2} \sqrt{2} f_x(x, y)^2 + \frac{1}{2} \sqrt{2} f_y(x, y)^2 + \frac{1}{2} \sqrt{2} \partial_x - \frac{1}{2} i \sqrt{2} f_x(x, y)^2 - \frac{1}{2} i \sqrt{2} f_y(x, y)^2 - \frac{1}{2} i \sqrt{2} \partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = 0$$

$$\lambda = 0$$

$$\begin{aligned}
\nu = & \sqrt{2}v^2 f_x(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial x)^2} f_x(x, y) + \sqrt{2}v^2 f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial x)^2} f_x(x, y) - (i-1) \sqrt{2}v^2 f_x(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{\partial x \partial y} f_x(x, y) \\
& - (i-1) \sqrt{2}v^2 f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{\partial x \partial y} f_x(x, y) + \sqrt{2}v^2 f_x(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) + \sqrt{2}v^2 f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) \\
& - (i-1) \sqrt{2}v^2 f_x(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) - (i-1) \sqrt{2}v^2 f_y(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) \\
& - i \sqrt{2}v^2 f_x(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) - i \sqrt{2}v^2 f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) - i \sqrt{2}v^2 f_x(x, y)^2 \frac{\partial}{\partial y} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) \\
& - i \sqrt{2}v^2 f_y(x, y)^2 \frac{\partial}{\partial y} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) + \sqrt{2}v^2 f_x(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial x)^2} f_y(x, y) + \sqrt{2}v^2 f_y(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial x)^2} f_y(x, y) \\
& - (i-1) \sqrt{2}v^2 f_x(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{\partial x \partial y} f_y(x, y) - (i-1) \sqrt{2}v^2 f_y(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{\partial x \partial y} f_y(x, y) \\
& + \sqrt{2}v^2 f_x(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) + \sqrt{2}v^2 f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) - (i-1) \sqrt{2}v^2 f_x(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y)
\end{aligned}$$

ν (continued)

$$\begin{aligned}
& - (i-1) \sqrt{2} v^2 f_y(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) - i \sqrt{2} v^2 f_x(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) - i \sqrt{2} v^2 f_y(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) \\
& - i \sqrt{2} v^2 f_x(x, y)^2 \frac{\partial}{\partial y} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) - i \sqrt{2} v^2 f_y(x, y)^2 \frac{\partial}{\partial y} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) + \sqrt{2} v^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial x)^2} f_x(x, y) \\
& - (i-1) \sqrt{2} v^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{\partial x \partial y} f_x(x, y) + \sqrt{2} v^2 \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) - (i-1) \sqrt{2} v^2 \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) - i \sqrt{2} v^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) \\
& - i \sqrt{2} v^2 \frac{\partial}{\partial y} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) + \sqrt{2} v^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial x)^2} f_y(x, y) - (i-1) \sqrt{2} v^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{\partial x \partial y} f_y(x, y) + \sqrt{2} v^2 \frac{\partial^2}{(\partial x)^2} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) \\
& - (i-1) \sqrt{2} v^2 \frac{\partial^2}{\partial x \partial y} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) - i \sqrt{2} v^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) - i \sqrt{2} v^2 \frac{\partial}{\partial y} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) + \frac{1}{2} \sqrt{2} f_x(x, y)^2 \frac{\partial}{\partial x} H^0(u, x, y) \\
& + \frac{1}{2} \sqrt{2} f_y(x, y)^2 \frac{\partial}{\partial x} H^0(u, x, y) - \frac{1}{2} i \sqrt{2} f_x(x, y)^2 \frac{\partial}{\partial y} H^0(u, x, y) - \frac{1}{2} i \sqrt{2} f_y(x, y)^2 \frac{\partial}{\partial y} H^0(u, x, y) + \frac{1}{2} \sqrt{2} \frac{\partial}{\partial x} H^0(u, x, y) - \frac{1}{2} i \sqrt{2} \frac{\partial}{\partial y} H^0(u, x, y)
\end{aligned}$$

$$\alpha = \frac{1}{2} \sqrt{2} f_x(x, y) \frac{\partial}{\partial x} f_x(x, y) - \frac{1}{2} i \sqrt{2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) + \frac{1}{2} \sqrt{2} f_y(x, y) \frac{\partial}{\partial x} f_y(x, y) - \frac{1}{2} i \sqrt{2} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y)$$

$$\beta = -\frac{1}{2} \sqrt{2} f_x(x, y) \frac{\partial}{\partial x} f_x(x, y) - \frac{1}{2} i \sqrt{2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) - \frac{1}{2} \sqrt{2} f_y(x, y) \frac{\partial}{\partial x} f_y(x, y) - \frac{1}{2} i \sqrt{2} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y)$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = -v \frac{\partial}{\partial x} f_x(x, y)^2 - 2v \frac{\partial}{\partial x} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) - v \frac{\partial}{\partial y} f_x(x, y)^2 - v \frac{\partial}{\partial x} f_y(x, y)^2 - 2v \frac{\partial}{\partial x} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) - v \frac{\partial}{\partial y} f_y(x, y)^2$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\begin{aligned}\Phi_{11} = & -\frac{1}{2} f_x(x, y)^2 \frac{\partial}{\partial x} f_x(x, y)^2 + \frac{1}{2} f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y)^2 + \frac{1}{2} f_x(x, y)^3 \frac{\partial^2}{(\partial x)^2} f_x(x, y) + \frac{1}{2} f_x(x, y) f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_x(x, y) - \frac{1}{2} f_x(x, y)^2 \frac{\partial}{\partial y} f_x(x, y)^2 \\ & + \frac{1}{2} f_y(x, y)^2 \frac{\partial}{\partial y} f_x(x, y)^2 + \frac{1}{2} f_x(x, y)^3 \frac{\partial^2}{(\partial y)^2} f_x(x, y) + \frac{1}{2} f_x(x, y) f_y(x, y)^2 \frac{\partial^2}{(\partial y)^2} f_x(x, y) - 2 f_x(x, y) f_y(x, y) \frac{\partial}{\partial x} f_x(x, y) \frac{\partial}{\partial x} f_y(x, y) + \frac{1}{2} f_x(x, y)^2 \frac{\partial}{\partial x} f_y(x, y)^2 \\ & - \frac{1}{2} f_y(x, y)^2 \frac{\partial}{\partial x} f_y(x, y)^2 + \frac{1}{2} f_x(x, y)^2 f_y(x, y) \frac{\partial^2}{(\partial x)^2} f_y(x, y) + \frac{1}{2} f_y(x, y)^3 \frac{\partial^2}{(\partial x)^2} f_y(x, y) - 2 f_x(x, y) f_y(x, y) \frac{\partial}{\partial y} f_x(x, y) \frac{\partial}{\partial y} f_y(x, y) + \frac{1}{2} f_x(x, y)^2 \frac{\partial}{\partial y} f_y(x, y)^2 \\ & - \frac{1}{2} f_y(x, y)^2 \frac{\partial}{\partial y} f_y(x, y)^2 + \frac{1}{2} f_x(x, y)^2 f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) + \frac{1}{2} f_y(x, y)^3 \frac{\partial^2}{(\partial y)^2} f_y(x, y) + \frac{\partial}{\partial x} f_x(x, y)^2 + \frac{1}{2} f_x(x, y) \frac{\partial^2}{(\partial x)^2} f_x(x, y) + \frac{\partial}{\partial x} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) \\ & + \frac{\partial}{\partial y} f_x(x, y)^2 + \frac{1}{2} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) + \frac{\partial}{\partial x} f_y(x, y)^2 + \frac{1}{2} f_y(x, y) \frac{\partial^2}{(\partial x)^2} f_y(x, y) + \frac{\partial}{\partial x} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) + \frac{\partial}{\partial y} f_y(x, y)^2 + \frac{1}{2} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y)\end{aligned}$$

$$\begin{aligned}\Phi_{12} = & -\sqrt{2} v f_x(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial x)^2} f_x(x, y) - \sqrt{2} v f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial x)^2} f_x(x, y) - (i+1) \sqrt{2} v f_x(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{\partial x \partial y} f_x(x, y) \\ & - (i+1) \sqrt{2} v f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{\partial x \partial y} f_x(x, y) - \sqrt{2} v f_x(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) - \sqrt{2} v f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) \\ & - (i+1) \sqrt{2} v f_x(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) - (i+1) \sqrt{2} v f_y(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) \\ & - i \sqrt{2} v f_x(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) - i \sqrt{2} v f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) - i \sqrt{2} v f_x(x, y)^2 \frac{\partial}{\partial y} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) \\ & - i \sqrt{2} v f_y(x, y)^2 \frac{\partial}{\partial y} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) - \sqrt{2} v f_x(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial x)^2} f_y(x, y) - \sqrt{2} v f_y(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial x)^2} f_y(x, y) \\ & - (i+1) \sqrt{2} v f_x(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{\partial x \partial y} f_y(x, y) - (i+1) \sqrt{2} v f_y(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{\partial x \partial y} f_y(x, y) - \sqrt{2} v f_x(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) \\ & - \sqrt{2} v f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y)\end{aligned}$$

Φ_{12} (continued)

$$\begin{aligned}
& -\sqrt{2}v f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) - (i+1) \sqrt{2}v f_x(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) - (i+1) \sqrt{2}v f_y(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) \\
& -i \sqrt{2}v f_x(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) - i \sqrt{2}v f_y(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) - i \sqrt{2}v f_x(x, y)^2 \frac{\partial}{\partial y} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) \\
& -i \sqrt{2}v f_y(x, y)^2 \frac{\partial}{\partial y} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) - \sqrt{2}v \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial x)^2} f_x(x, y) - (i+1) \sqrt{2}v \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{\partial x \partial y} f_x(x, y) \\
& -\sqrt{2}v \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) - (i+1) \sqrt{2}v \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) - i \sqrt{2}v \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) \\
& -i \sqrt{2}v \frac{\partial}{\partial y} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) - \sqrt{2}v \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial x)^2} f_y(x, y) - (i+1) \sqrt{2}v \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{\partial x \partial y} f_y(x, y) - \sqrt{2}v \frac{\partial^2}{(\partial x)^2} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) \\
& - (i+1) \sqrt{2}v \frac{\partial^2}{\partial x \partial y} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) - i \sqrt{2}v \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) - i \sqrt{2}v \frac{\partial}{\partial y} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y)
\end{aligned}$$

$\Phi_{20} = 0$

$$\begin{aligned}
\Phi_{21} = & -\sqrt{2}v f_x(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial x)^2} f_x(x, y) - \sqrt{2}v f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial x)^2} f_x(x, y) + (i-1) \sqrt{2}v f_x(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{\partial x \partial y} f_x(x, y) \\
& + (i-1) \sqrt{2}v f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{\partial x \partial y} f_x(x, y) - \sqrt{2}v f_x(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) - \sqrt{2}v f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) \\
& + (i-1) \sqrt{2}v f_x(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) + (i-1) \sqrt{2}v f_y(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) \\
& + i \sqrt{2}v f_x(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) + i \sqrt{2}v f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) + i \sqrt{2}v f_x(x, y)^2 \frac{\partial}{\partial y} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) \\
& + i \sqrt{2}v f_y(x, y)^2 \frac{\partial}{\partial y} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) - \sqrt{2}v f_x(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial x)^2} f_y(x, y) - \sqrt{2}v f_y(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial x)^2} f_y(x, y) \\
& + (i-1) \sqrt{2}v f_x(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{\partial x \partial y} f_y(x, y) + (i-1) \sqrt{2}v f_y(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{\partial x \partial y} f_y(x, y) - \sqrt{2}v f_x(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y)
\end{aligned}$$

$$\begin{aligned}
& -\sqrt{2}v f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) + (i-1) \sqrt{2}v f_x(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) + (i-1) \sqrt{2}v f_y(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) \\
& + i \sqrt{2}v f_x(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) + i \sqrt{2}v f_y(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) + i \sqrt{2}v f_x(x, y)^2 \frac{\partial}{\partial y} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) \\
& + i \sqrt{2}v f_y(x, y)^2 \frac{\partial}{\partial y} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) - \sqrt{2}v \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial x)^2} f_x(x, y) + (i-1) \sqrt{2}v \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{\partial x \partial y} f_x(x, y) \\
& - \sqrt{2}v \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) + (i-1) \sqrt{2}v \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) + i \sqrt{2}v \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) \\
& + i \sqrt{2}v \frac{\partial}{\partial y} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) - \sqrt{2}v \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial x)^2} f_y(x, y) + (i-1) \sqrt{2}v \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{\partial x \partial y} f_y(x, y) - \sqrt{2}v \frac{\partial^2}{(\partial x)^2} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) \\
& + (i-1) \sqrt{2}v \frac{\partial^2}{\partial x \partial y} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) + i \sqrt{2}v \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) + i \sqrt{2}v \frac{\partial}{\partial y} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y)
\end{aligned}$$

7

Φ_{22} (continued)

$$\begin{aligned}
& + 2v^2 f_y(x, y)^4 \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) + v^2 f_x(x, y)^4 \frac{\partial^2}{(\partial y)^2} f_x(x, y)^2 + 2v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial^2}{(\partial y)^2} f_x(x, y)^2 \\
& + v^2 f_y(x, y)^4 \frac{\partial^2}{(\partial y)^2} f_x(x, y)^2 + v^2 f_x(x, y)^4 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^3}{(\partial y)^3} f_x(x, y) + 2v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^3}{(\partial y)^3} f_x(x, y) \\
& + v^2 f_y(x, y)^4 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^3}{(\partial y)^3} f_x(x, y) + v^2 f_x(x, y)^4 \frac{\partial}{\partial y} f_x(x, y) \frac{\partial^3}{(\partial y)^3} f_x(x, y) + 2v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial}{\partial y} f_x(x, y) \frac{\partial^3}{(\partial y)^3} f_x(x, y) \\
& + v^2 f_y(x, y)^4 \frac{\partial}{\partial y} f_x(x, y) \frac{\partial^3}{(\partial y)^3} f_x(x, y) + v^2 f_x(x, y)^4 \frac{\partial^2}{(\partial x)^2} f_y(x, y)^2 + 2v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_y(x, y)^2 \\
& + v^2 f_y(x, y)^4 \frac{\partial^2}{(\partial x)^2} f_y(x, y)^2 + v^2 f_x(x, y)^4 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^3}{(\partial x)^3} f_y(x, y) + 2v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^3}{(\partial x)^3} f_y(x, y) \\
& + v^2 f_y(x, y)^4 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^3}{(\partial x)^3} f_y(x, y) + v^2 f_x(x, y)^4 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^3}{(\partial x)^2 \partial y} f_y(x, y) + 2v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^3}{(\partial x)^2 \partial y} f_y(x, y) \\
& + v^2 f_y(x, y)^4 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^3}{(\partial x)^2 \partial y} f_y(x, y) + 2v^2 f_x(x, y)^4 \frac{\partial^2}{(\partial x)^2} f_y(x, y) \frac{\partial^2}{\partial x \partial y} f_y(x, y) + 4v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_y(x, y) \frac{\partial^2}{\partial x \partial y} f_y(x, y) \\
& + 2v^2 f_y(x, y)^4 \frac{\partial^2}{(\partial x)^2} f_y(x, y) \frac{\partial^2}{\partial x \partial y} f_y(x, y) + 2v^2 f_x(x, y)^4 \frac{\partial^2}{\partial x \partial y} f_y(x, y)^2 + 4v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_y(x, y)^2 + 2v^2 f_y(x, y)^4 \frac{\partial^2}{\partial x \partial y} f_y(x, y)^2 \\
& + v^2 f_x(x, y)^4 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^3}{\partial x (\partial y)^2} f_y(x, y) + 2v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^3}{\partial x (\partial y)^2} f_y(x, y) + v^2 f_y(x, y)^4 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^3}{\partial x (\partial y)^2} f_y(x, y) \\
& + v^2 f_x(x, y)^4 \frac{\partial^3}{(\partial x)^3} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) + 2v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial^3}{(\partial x)^3} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) + v^2 f_y(x, y)^4 \frac{\partial^3}{(\partial x)^3} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) + \dots
\end{aligned}$$

$$\begin{aligned}
\Lambda = & -\frac{1}{6} f_x(x, y)^2 \frac{\partial}{\partial x} f_x(x, y)^2 + \frac{1}{6} f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y)^2 + \frac{1}{6} f_x(x, y)^3 \frac{\partial^2}{(\partial x)^2} f_x(x, y) + \frac{1}{6} f_x(x, y) f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_x(x, y) - \frac{1}{6} f_x(x, y)^2 \frac{\partial}{\partial y} f_x(x, y)^2 \\
& + \frac{1}{6} f_y(x, y)^2 \frac{\partial}{\partial y} f_x(x, y)^2 + \frac{1}{6} f_x(x, y)^3 \frac{\partial^2}{(\partial y)^2} f_x(x, y) + \frac{1}{6} f_x(x, y) f_y(x, y)^2 \frac{\partial^2}{(\partial y)^2} f_x(x, y) - \frac{2}{3} f_x(x, y) f_y(x, y) \frac{\partial}{\partial x} f_x(x, y) \frac{\partial}{\partial x} f_y(x, y) \\
& + \frac{1}{6} f_x(x, y)^2 \frac{\partial}{\partial x} f_y(x, y)^2 - \frac{1}{6} f_y(x, y)^2 \frac{\partial}{\partial x} f_y(x, y)^2 + \frac{1}{6} f_x(x, y)^2 f_y(x, y) \frac{\partial^2}{(\partial x)^2} f_y(x, y) + \frac{1}{6} f_y(x, y)^3 \frac{\partial^2}{(\partial x)^2} f_y(x, y) - \frac{2}{3} f_x(x, y) f_y(x, y) \frac{\partial}{\partial y} f_x(x, y) \frac{\partial}{\partial y} f_y(x, y) \\
& + \frac{1}{6} f_x(x, y)^2 \frac{\partial}{\partial y} f_y(x, y)^2 - \frac{1}{6} f_y(x, y)^2 \frac{\partial}{\partial y} f_y(x, y)^2 + \frac{1}{6} f_x(x, y)^2 f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) + \frac{1}{6} f_y(x, y)^3 \frac{\partial^2}{(\partial y)^2} f_y(x, y) + \frac{1}{6} f_x(x, y) \frac{\partial^2}{(\partial x)^2} f_x(x, y) \\
& - \frac{1}{3} \frac{\partial}{\partial x} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) + \frac{1}{6} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) + \frac{1}{6} f_y(x, y) \frac{\partial^2}{(\partial x)^2} f_y(x, y) - \frac{1}{3} \frac{\partial}{\partial x} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) + \frac{1}{6} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y)
\end{aligned}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\begin{aligned}\Psi_2 = & \frac{1}{3} f_x(x, y)^2 \frac{\partial}{\partial x} f_x(x, y)^2 - \frac{1}{3} f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y)^2 - \frac{1}{3} f_x(x, y)^3 \frac{\partial^2}{(\partial x)^2} f_x(x, y) - \frac{1}{3} f_x(x, y) f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_x(x, y) + \frac{1}{3} f_x(x, y)^2 \frac{\partial}{\partial y} f_x(x, y)^2 \\ & - \frac{1}{3} f_y(x, y)^2 \frac{\partial}{\partial y} f_x(x, y)^2 - \frac{1}{3} f_x(x, y)^3 \frac{\partial^2}{(\partial y)^2} f_x(x, y) - \frac{1}{3} f_x(x, y) f_y(x, y)^2 \frac{\partial^2}{(\partial y)^2} f_x(x, y) + \frac{4}{3} f_x(x, y) f_y(x, y) \frac{\partial}{\partial x} f_x(x, y) \frac{\partial}{\partial x} f_y(x, y) \\ & - \frac{1}{3} f_x(x, y)^2 \frac{\partial}{\partial x} f_y(x, y)^2 + \frac{1}{3} f_y(x, y)^2 \frac{\partial}{\partial x} f_y(x, y)^2 - \frac{1}{3} f_x(x, y)^2 f_y(x, y) \frac{\partial^2}{(\partial x)^2} f_y(x, y) - \frac{1}{3} f_y(x, y)^3 \frac{\partial^2}{(\partial x)^2} f_y(x, y) + \frac{4}{3} f_x(x, y) f_y(x, y) \frac{\partial}{\partial y} f_x(x, y) \frac{\partial}{\partial y} f_y(x, y) \\ & - \frac{1}{3} f_x(x, y)^2 \frac{\partial}{\partial y} f_y(x, y)^2 + \frac{1}{3} f_y(x, y)^2 \frac{\partial}{\partial y} f_y(x, y)^2 - \frac{1}{3} f_x(x, y)^2 f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) - \frac{1}{3} f_y(x, y)^3 \frac{\partial^2}{(\partial y)^2} f_y(x, y) - \frac{1}{3} f_x(x, y) \frac{\partial^2}{(\partial x)^2} f_x(x, y) \\ & + \frac{2}{3} \frac{\partial}{\partial x} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) - \frac{1}{3} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) - \frac{1}{3} f_y(x, y) \frac{\partial^2}{(\partial x)^2} f_y(x, y) + \frac{2}{3} \frac{\partial}{\partial x} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) - \frac{1}{3} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y)\end{aligned}$$

$$\begin{aligned}\Psi_3 = & -\sqrt{2}v f_x(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial x)^2} f_x(x, y) - \sqrt{2}v f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial x)^2} f_x(x, y) + (i-1) \sqrt{2}v f_x(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{\partial x \partial y} f_x(x, y) \\ & + (i-1) \sqrt{2}v f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{\partial x \partial y} f_x(x, y) - \sqrt{2}v f_x(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) - \sqrt{2}v f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) \\ & + (i-1) \sqrt{2}v f_x(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) + (i-1) \sqrt{2}v f_y(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) \\ & + i \sqrt{2}v f_x(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) + i \sqrt{2}v f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) + i \sqrt{2}v f_x(x, y)^2 \frac{\partial}{\partial y} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) \\ & + i \sqrt{2}v f_y(x, y)^2 \frac{\partial}{\partial y} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) - \sqrt{2}v f_x(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial x)^2} f_y(x, y) - \sqrt{2}v f_y(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial x)^2} f_y(x, y) \\ & + (i-1) \sqrt{2}v f_x(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{\partial x \partial y} f_y(x, y) + (i-1) \sqrt{2}v f_y(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{\partial x \partial y} f_y(x, y) - \sqrt{2}v f_x(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) \\ & - \sqrt{2}v f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y)\end{aligned}$$

Ψ_3 (continued)

$$\begin{aligned}
& -\sqrt{2}v f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) + (i-1) \sqrt{2}v f_x(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) + (i-1) \sqrt{2}v f_y(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) \\
& + i \sqrt{2}v f_x(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) + i \sqrt{2}v f_y(x, y)^2 \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) + i \sqrt{2}v f_x(x, y)^2 \frac{\partial}{\partial y} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) \\
& + i \sqrt{2}v f_y(x, y)^2 \frac{\partial}{\partial y} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) - \sqrt{2}v \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial x)^2} f_x(x, y) + (i-1) \sqrt{2}v \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{\partial x \partial y} f_x(x, y) \\
& - \sqrt{2}v \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) + (i-1) \sqrt{2}v \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) + i \sqrt{2}v \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) \\
& + i \sqrt{2}v \frac{\partial}{\partial y} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) - \sqrt{2}v \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial x)^2} f_y(x, y) + (i-1) \sqrt{2}v \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{\partial x \partial y} f_y(x, y) - \sqrt{2}v \frac{\partial^2}{(\partial x)^2} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) \\
& + (i-1) \sqrt{2}v \frac{\partial^2}{\partial x \partial y} f_y(x, y) \frac{\partial}{\partial y} f_y(x, y) + i \sqrt{2}v \frac{\partial}{\partial x} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y) + i \sqrt{2}v \frac{\partial}{\partial y} f_y(x, y) \frac{\partial^2}{(\partial y)^2} f_y(x, y)
\end{aligned}$$

$$\begin{aligned}
\Psi_4 = & 4v^2 f_x(x, y)^3 \frac{\partial}{\partial x} f_x(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_x(x, y) + 4v^2 f_x(x, y) f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_x(x, y) + v^2 f_x(x, y)^4 \frac{\partial^2}{(\partial x)^2} f_x(x, y)^2 + 2v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_x(x, y)^2 \\
& + v^2 f_y(x, y)^4 \frac{\partial^2}{(\partial x)^2} f_x(x, y)^2 + v^2 f_x(x, y)^4 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^3}{(\partial x)^3} f_x(x, y) + 2v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^3}{(\partial x)^3} f_x(x, y) + v^2 f_y(x, y)^4 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^3}{(\partial x)^3} f_x(x, y) \\
& - (2i-1) v^2 f_x(x, y)^4 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^3}{(\partial x)^2 \partial y} f_x(x, y) - (4i-2) v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^3}{(\partial x)^2 \partial y} f_x(x, y) - (2i-1) v^2 f_y(x, y)^4 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^3}{(\partial x)^2 \partial y} f_x(x, y) \\
& - (4i-4) v^2 f_x(x, y)^3 \frac{\partial}{\partial x} f_x(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_x(x, y) - (4i-4) v^2 f_x(x, y) f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_x(x, y) - (2i-2) v^2 f_x(x, y)^4 \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial^2}{\partial x \partial y} f_x(x, y) \\
& - (4i-4) v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial^2}{\partial x \partial y} f_x(x, y) - (2i-2) v^2 f_y(x, y)^4 \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial^2}{\partial x \partial y} f_x(x, y)^2 \\
& - 4i v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_x(x, y)^2 - 2i v^2 f_y(x, y)^4 \frac{\partial^2}{\partial x \partial y} f_x(x, y)^2 - (2i+1) v^2 f_x(x, y)^4 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^3}{\partial x (\partial y)^2} f_x(x, y) \\
& - (4i+2) v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^3}{\partial x (\partial y)^2} f_x(x, y) - (2i+1) v^2 f_y(x, y)^4 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^3}{\partial x (\partial y)^2} f_x(x, y) \\
& - (4i-4) v^2 f_x(x, y)^3 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) - (4i-4) v^2 f_x(x, y) f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) \\
& + v^2 f_x(x, y)^4 \frac{\partial^3}{(\partial x)^3} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) + 2v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial^3}{(\partial x)^3} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) + v^2 f_y(x, y)^4 \frac{\partial^3}{(\partial x)^3} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y)
\end{aligned}$$

Ψ_4 (continued)

$$\begin{aligned}
& - (2i - 1) v^2 f_x(x, y)^4 \frac{\partial^3}{(\partial x)^2 \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) - (4i - 2) v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial^3}{(\partial x)^2 \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) \\
& - (2i - 1) v^2 f_y(x, y)^4 \frac{\partial^3}{(\partial x)^2 \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) - 8i v^2 f_x(x, y)^3 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) \\
& - 8i v^2 f_x(x, y) f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) - (2i + 1) v^2 f_x(x, y)^4 \frac{\partial^3}{\partial x (\partial y)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) \\
& - (4i + 2) v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial^3}{\partial x (\partial y)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) - (2i + 1) v^2 f_y(x, y)^4 \frac{\partial^3}{\partial x (\partial y)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) - 4i v^2 f_x(x, y)^3 \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y)^2 \\
& - 4i v^2 f_x(x, y) f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y)^2 - (4i + 4) v^2 f_x(x, y)^3 \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y)^2 - (4i + 4) v^2 f_x(x, y) f_y(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y)^2 \\
& - 4i v^2 f_x(x, y)^3 \frac{\partial}{\partial x} f_x(x, y)^2 \frac{\partial^2}{(\partial y)^2} f_x(x, y) - 4i v^2 f_x(x, y) f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y)^2 \frac{\partial^2}{(\partial y)^2} f_x(x, y) - 2i v^2 f_x(x, y)^4 \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) \\
& - 4i v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) - 2i v^2 f_y(x, y)^4 \frac{\partial^2}{(\partial x)^2} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) - (2i + 2) v^2 f_x(x, y)^4 \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) \\
& - (4i + 4) v^2 f_x(x, y)^2 f_y(x, y)^2 \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) - (2i + 2) v^2 f_y(x, y)^4 \frac{\partial^2}{\partial x \partial y} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) \\
& - (4i + 4) v^2 f_x(x, y)^3 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) - (4i + 4) v^2 f_x(x, y) f_y(x, y)^2 \frac{\partial}{\partial x} f_x(x, y) \frac{\partial}{\partial y} f_x(x, y) \frac{\partial^2}{(\partial y)^2} f_x(x, y) \\
& - 4 v^2 f_x(x, y)^3 \frac{\partial}{\partial y} f_x(x, y)^2 \frac{\partial^2}{(\partial y)^2} f_x(x, y) - 4 v^2 f_x(x, y) f_y(x, y)^2 \frac{\partial}{\partial y} f_x(x, y)^2 \frac{\partial^2}{(\partial y)^2} f_x(x, y) + \dots
\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"